Identifying Promoter Sequences using Neural Networks

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Introduction

Promoters are DNA sequences that regulate gene expression by serving as binding sites for RNA polymerase and transcription factors, initiating transcription[i]. Identifying promoters is crucial for understanding gene regulation, as they play a critical role in controlling when and where genes are transcribed. In their 2017 PLOS paper[ii], Solovyev and Umarov trained CNNs to classify given sequences from various organisms into promoters and non-promoters. In this project we aimed to reproduce their results, try to improve upon them, and explore possible interpretations.

It is experimentally possible to locate promoters in the genome using methods like ATAC-seq, which find histone modifications that indicate presence of promoters[iii]. However, this data is not always available and is dependent on the tissue. Thus, being able to predict the location of a promoter just based on the DNA sequence is useful and might also help highlight biological effects such as protein binding sites.

Data

Our data came from two sources, the PLOS paper and the ENSEMBL genome browser.

PLOS:

The data used in the paper consists of FASTA files containing promoter and non-promoter sequences. The promoter sequences were found using the EPD dataset[iv], taking the [–200 : +50] range relative to the TSS for eukaryotes and [-60 : +20] for prokaryotes. The non-promoter sequences were randomly selected from coding segments downstream of the promoters and then translated by reverse complement to the non-coding strand.

Organism	Promoter sequences	Non-promoter sequences
Human	19,811	27,731
Mouse	16,283	24,822
Arabidopsis	1,497	2,879
E. Coli	839	3000

Table 1 – data from the PLOS paper

ENSEMBL:

The ENSEMBL database is a genome database with various genetic data, including annotation of regulatory features, such as promoters. The promoter annotation is based on histone modifications from Dnase-seq and ATAC-seq and covers the whole human genome.[v]

Using the ENSEMBL database and the pyranges library, we downloaded promoter annotation regions from the human genome and padded the sequences to a sequence of length 2000. To maintain consistency with the PLOS data, we limited the sequences to promoters up to 251 bp. We wanted sequences that start before the promoter and end after it, so we padded the promoters upstream and downstream to get sequences 2000 bp long. The distribution of the padding before and after was random. For each sequence, we created a corresponding sequence of annotations, with P for bases originally labeled “promoter” and N for bases added by the padding.

Methods

We trained three types of models – HMM, K-mer based SVM, and CNNs.

HMM:

Hidden Markov Models (HMMs) are models that assume the system may be described as being in one of a finite set of states. The system may then transition between states according to a given probability distribution (transition probabilities), where each transition depends entirely on the current state (Markov property). Each state in turn is associated with the probability to

emit any of a set of possible observations (emission probabilities). The model assumes that we cannot observe the hidden states, but rather only the emissions. [vi]

We employed an HMM with two hidden states, promoter (P) and non-promoter (N). The emission probabilities are dependent on the previous two nucleotides in the sequence:

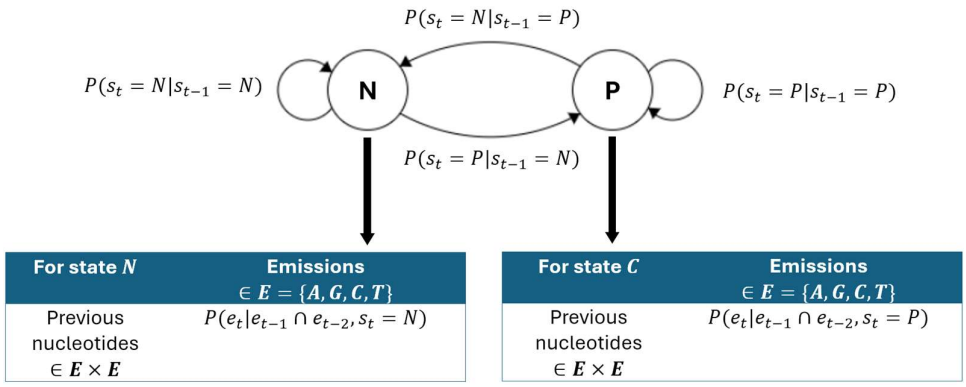

Figure 1 – HMM model

In order to learn the parameters of the model, we calculated maximum likelihood estimators for the transition and emission probabilities by averaging the number of transitions and emissions of each type over all the transitions and emissions, respectively:

$$\hat{P}_{MLE}(s_t = S|s_{t-1} = S') = \Sigma_{i=2}^{n}\left(\frac{\#(s_i = S \cap s_{i-1} = S')}{\#(s_{i-1} = S')}\right)$$

$$\hat{P}_{MLE}(e_t|e_{t-1} \cap e_{t-2}, s_t = S) = \Sigma_{i=3}^{n}\left(\frac{\#(e_i = E \cap e_{i-1} = E' \cap e_{i-2} = E'' \cap s_i = S)}{\#(e_{i-1} = E' \cap e_{i-2} = E'' \cap s_i = S)}\right)$$

K-mer based SVM:

The k-mer method analyzes DNA sequences by breaking them into overlapping subsequences of length k, allowing for frequency-based comparison between different sequence types. SVMs classify data by finding the optimal hyperplane that maximizes the margin between different classes, and they can use kernel functions to handle complex, non-linearly separable data.

For each sequence, we computed the frequency of k-mers by counting the occurrence of every possible k-mer in the sequence. Specifically, we focused on 4-mers and 7-mers to capture short and mid-length sequence patterns relevant to promoter regions. Each sequence was converted into a count vector of size 4^k, representing the frequency of all possible k-mers in the sequence, which was then normalized by the total k-mer count within each sequence.

To identify k-mers that exhibit significant differences between promoter and non-promoter regions, we calculated the difference in normalized counts for each k-mer across the two sets. The difference in normalized counts was then sorted by the absolute magnitude of the

difference, which helped identify the k-mers that showed the largest disparity between the two regions.

We then trained an SVM using K-mer-based features to classify DNA sequences as promoters or non-promoters, with the default scikit-learn kernel. Separate analyses were performed for the human genome (non-TATA promoters) and the Arabidopsis genome (TATA promoters), allowing for organism-specific insights into promoter classification. We used the data from the PLOS paper, split into training (70%) and testing (30%) sets.

CNN:

Convolutional Neural Networks (CNNs) are feed forward neural networks that typically utilize alternating convolutional and pooling layers. Each convolutional layer consists of a set of filters applied to the data by discrete convolution, then usually passed through a non-linearity. Pooling layers merge the results from neighboring regions of the input into a single value, often computing the maximum value. These layers are usually followed by fully connected layers.[vii]

CNNs are suitable for our task since they do well at identifying local patterns, even if their location shifts.[vii] This spatial invariance is beneficial in promoter detection because regulatory motifs may appear at varying positions within DNA sequences.

Following the PLOS paper, we implemented the following architecture:

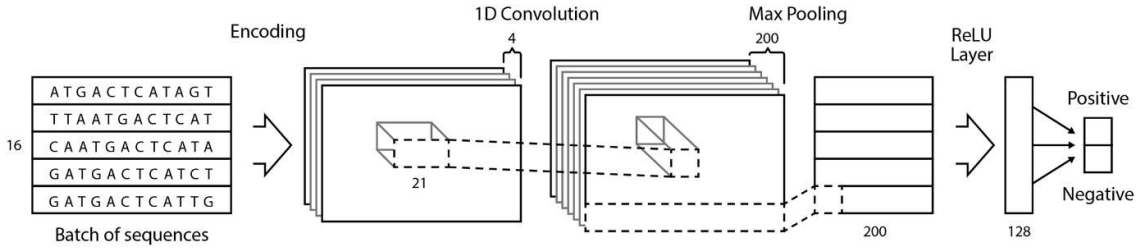

Figure 2 - CNN Architecture

Specifically, the network consists of one convolutional layer, one Max-Pool layer, one fully connected ReLU layer, and an output layer with sigmoid activation. The network parameters change between organisms as follows:

Organism	Convolution filter size	Number of Filters	MaxPool kernel size
Human	21	300	231
Mouse	21	250	2
Arabidopsis	21	200	4
E. Coli	21	150	12

Table 2 - Model Specification for Each Organism

We also implemented a simpler model consisting of 10 filters of size 7, with a Max-Pool kernel of size 2, and a more complex model with 3 convolutional layers (12 filters of size 7, 32×5, 64×3, each with pooling with a kernel of size 2) followed by a fully connected layer.

For the training of all the models we used the PLOS data, training on 72% of the data, validating and calibrating hyper-parameters using 8%, and tested our trained model on the remaining 20%, as was done in the paper.

Extracting Motifs from Filters

To understand motifs from the model, we extracted the kernels of each filter in the first convolution layer, which was of dimensions $4\ x\ 7$. We then converted to a PMW W using the SoftMax function with a size factor of 3 across the columns:

$$n_{i,j} = \frac{e^{3W_{i,j}}}{\sum_{k=0}^{3} e^{3W_{i,k}}}$$

Sequence logos were then created using the logomaker python package.

Window masking

Like the PLOS paper, we searched for functional sites that are defined by a location relative to the TSS (such as the TATA-BOX). We ran a sliding window in the size of our CNN's kernel (7 bases long) through the test set and for each window start position, masked the subsequence (by altering the original subsequence with zeros) and calculated the sensitivity of the model over all the masked sequences.

We then plotted the model's sensitivity by the start position of the masked sequence and looked for positions in which masking significantly reduced the sensitivity. We conclude that these positions include motifs that were important for our model's learning efficiency. Hopefully, those motifs are also biologically important sequences (such as transcription factors binding sites) that we would anticipate finding in promoters.

Model performance on new data

To see if the CNN is able to predict promoter boundaries on longer sequences from other datasets, we used the 2000 base sequences produced by ENSEMBL and ran across it using a 251-base window, calculating the model probability in each window.

Results

CNNs

Full models:

Organism	Accuracy	Specificity	Sensitivity	CC
Human	0.93	0.95	0.90	0.86
Mouse	0.92	0.94	0.88	0.82
Arabidopsis	0.98	0.97	0.99	0.95
E. Coli	0.94	0.97	0.85	0.83

Small models:

Organism	Accuracy	Specificity	Sensitivity	CC
Human	0.87	0.91	0.80	0.73
Mouse	0.92	0.95	0.87	0.83
Arabidopsis	0.96	0.95	0.96	0.91
E. Coli	0.89	0.94	0.74	0.69

Expanded model – 3 convolution layers:

Organism	Accuracy	Specificity	Sensitivity	CC
Human	0.88	0.97	0.76	0.77

Table 3 – Results of each model on test set.

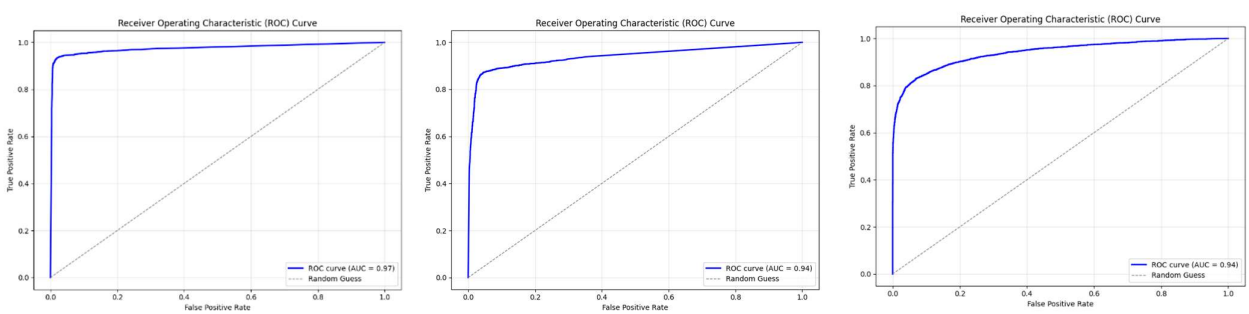

Figure 3 - ROC curves of CNN models (from left to right), Human full model (AUC 0.97), Human small model (AUC 0.94), Human expanded model (AUC 0.94)

HMMs

The HMM model's performance was poor, with minimal improvement over random assignment of labels. This implies that first or second order dependencies on the sequence are insufficient to accurately identify promoter regions.

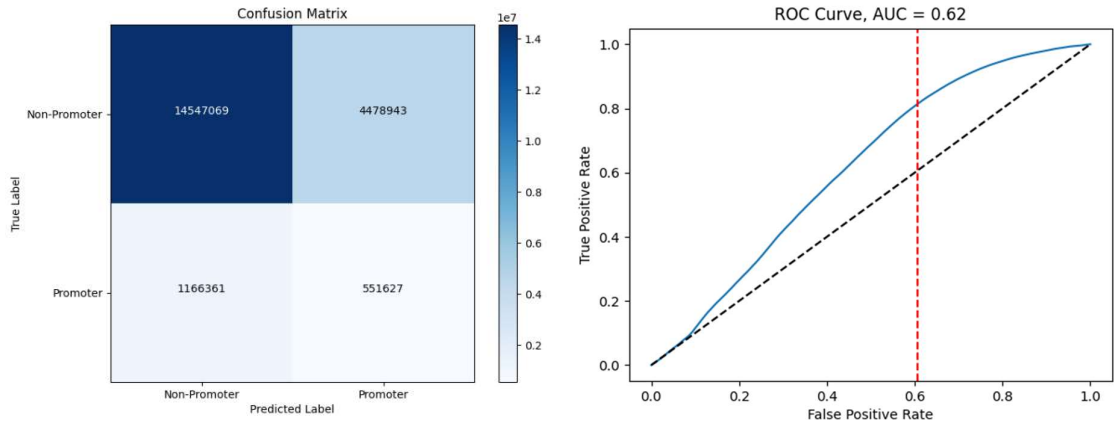

Figure 4 - HMM model results

K-mers

After extracting 4-mers from the data examined the k-mers with greatest difference between the promoter and non-promoter sequences. K-mers such as "GGGG," "GGGC," "CCGG," and others, which are rich in G and C, exhibited significantly higher frequencies in promoter sequences.

We then tried to detect a promotor sequence using an SVM with the k-mers as features with results shown below:

Organism	K	Accuracy	Specificity	Sensitivity	CC
Human	4	0.84	0.92	0.73	0.67
Arabidopsis	4	0.88	0.89	0.86	0.74
Arabidopsis	7	0.88	0.92	0.81	0.74

Table 4 - K-mer SVM model metrics on different datasets and K parameters

K-mer Feature Importance Aligns with Known Biology

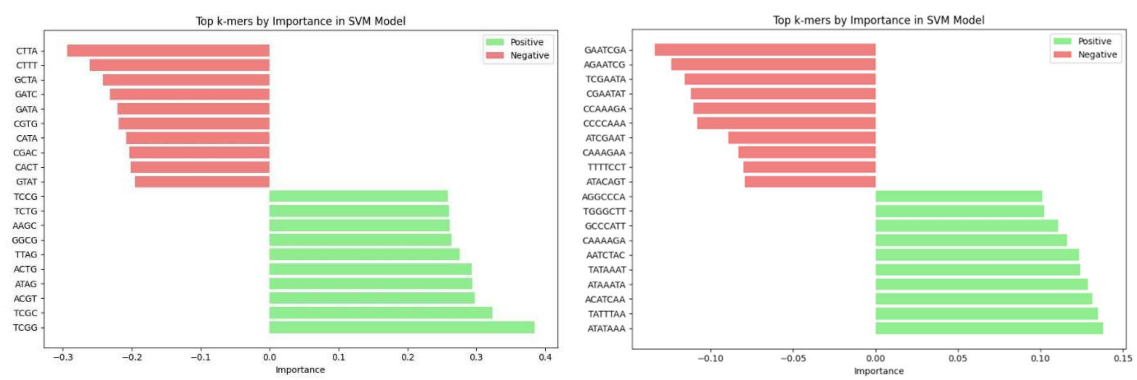

Figure 5 - SVM feature importance of k-mer model train on Arabidopsis with K=4 (left) and K=8 (right)

Analyzing the SVM model's feature weights revealed the most influential k-mers for promoter classification. Among the top k-mers for k=4, we observed that CpG-rich sequences are common, such as TCGG and TCGC, are among the most positively weighted k-mers associated with promoter regions. This observation aligns with the known biological characteristic of CpG islands being more frequent in promoters[viii] , as they play a crucial role in gene regulation and transcription initiation.

Among the top k-mers for k=7, we observed several sequences resembling the TATA box, a well-known core promoter element. For instance, TATAAAT and ATATAAA are variations of the TATA box sequence that are essential for the binding of transcription factors like TATA-binding protein (TBP)[ix].

Model performance on new data

When predicting on longer sequences (see methods) we found cases where the models (CNN, SVM with K-mer) assigned a higher probability once the window entered the promoter region, however this was generally not the case, and it often missed entirely - both models produced low precision and recall scores (<0.25) on the dataset. This might indicate that our model is overfitted on the PLOS data, which may not accurately represent the whole genome.

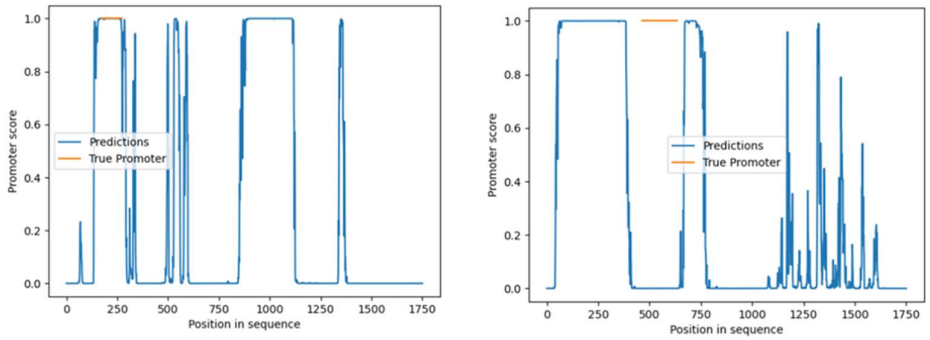

Figure 6 – CNN Model predictions on longer sequences (human data). Each point is the model probability for the window starting from this position to be a promoter, with the orange line being the actual bounderies of the promoter as found in ensembl. The left graph seems to detect a promoter whilst the right one misses entirely.

Model Interpretation using Window Masking and Kernel Visualization

As described in the methods, we ran a window mask and extracted sequence logos of the kernels learned by our small CNN model (10 kernels of size 7).

We ran the masking windows analysis over Arabidopsis and Mouse promoters. The Arabidopsis promoters in the data contains TATA boxes, while the Mouse promoters doesn't. In both runs we saw a significant decrease in the model's sensitivity in specific positions prior to the TSS. In Arabidopsis these positions were around 166-170, approximately 30 bases upstream of the TSS,

the exact TATA box location described in literaturex. Mouse data had a similar drop around the TSS. As a follow up, we extracted a sequence logo from the model's kernels. Arabidopsis data showed strong motifs of "TATA" variations in a few kernels, strengthening our conclusion.

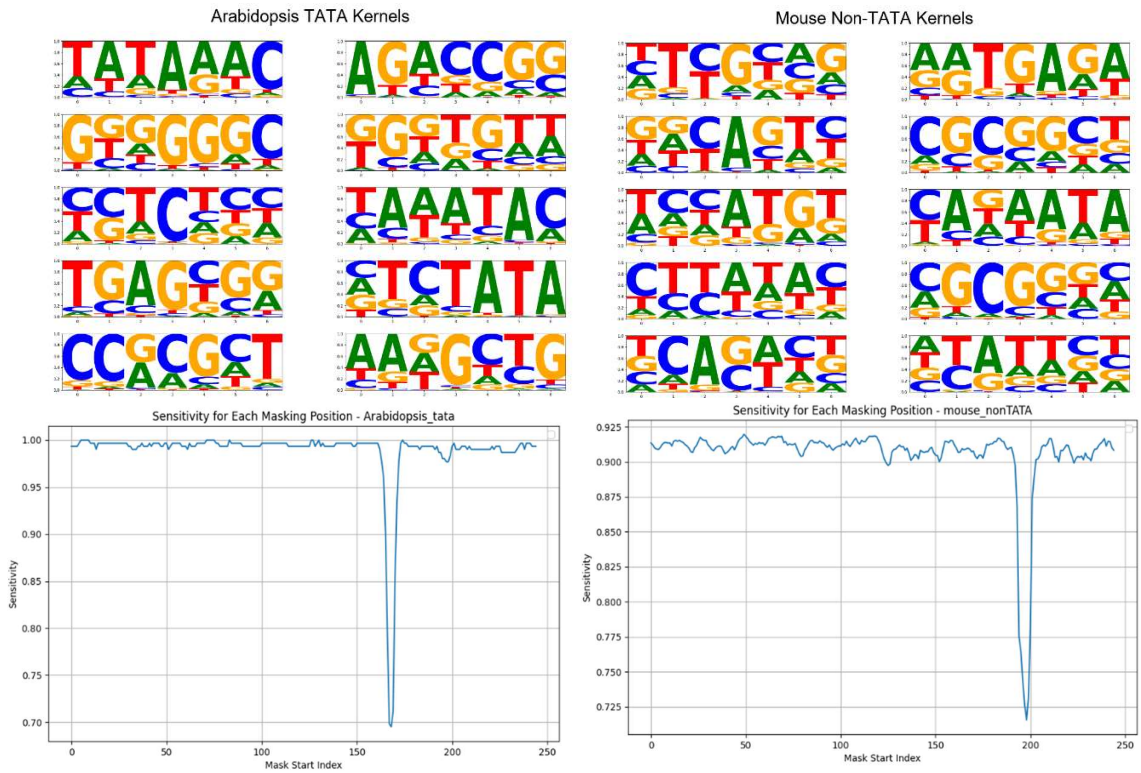

Figure 7 - Kernel Sequence Logos and Window Masking for Arabidopsis (left) and Mouse (right). Masking is based on the PLOS paper data, which has the TSS at position 200.

Discussion

Our k-mer based SVM had 0.84 accuracy on the human test set, not too far from the CNN peak performance (accuracy 0.93), suggesting that promoters have unique segments of specific sequences, and that training a CNN may be redundant. Most of the ability to distinguish between the promoter and non-promoter sequences can be explained by a few well-defined sequences, and not by complicated relations between many elements in the sequence.

This discovery seemed overly simplistic, so we evaluated our classifiers "in the wild" as explained in detail above and got poor results from both. Therefore, we assume that the data we worked with from the PLOS paper is biased and is not general enough for training a classifier that can identify promoter sequences genome wide.

This bias may be due to the fact that all its negatives are derived from reverse complements of coding regions, and not from random areas in the genome. It may also be due to general overfitting on the training data, although we tried to mitigate overfit by using simpler models.

Either way, training the models on better curated data (such as from the ENSEMBL dataset) may produce better results.

The k-mer analysis highlighted enrichment of G and C-rich k-mers in promoter regions, which aligns with known patterns of sequence motifs involved in gene regulation.[viii] These findings suggest that specific G and C-rich motifs might play a role in the recognition of promoter sequences or in the binding of transcription factors. Further investigation into the biological significance of these motifs could provide insights into gene regulation and the mechanisms underlying promoter function.

Between the different models, we encountered a common tradeoff between the performance and interpretability. Specifically, our best performing CNN architecture used 300 filters in its convolution layer, which can be interpreted as 300 questions that we ask, and using some combination of these answers allows the model to predict if the sequence is a promoter. Using a much slimmer architecture, with only 10 filters, the results were slightly poorer, but we eventually succeeded in interpreting the kernels manually and even identify known TATA-Box sequences. This method with 300 filters would take much greater effort, and may not be possible with more layers.

Conclusion:

In our project, we successfully reproduced the results we set out to reproduce. We did not succeed in improving on the paper's model, however in our attempts we found that the results generalize poorly with real-world data, suggesting that the training data may not be representative of the whole genome. In addition, we achieved similar results with significantly simpler models, allowing much more interpretability and uncovering known biological properties.

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